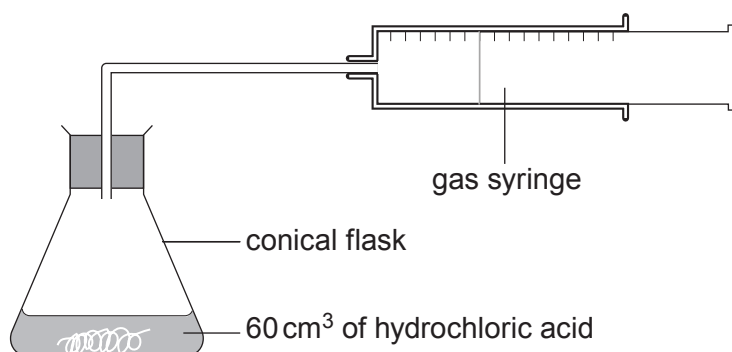


4. Magnesium ribbon was reacted with excess dilute hydrochloric acid. The volume of hydrogen produced during the reaction was measured over time using a gas syringe. The experiment was carried out at a temperature of 20°C .



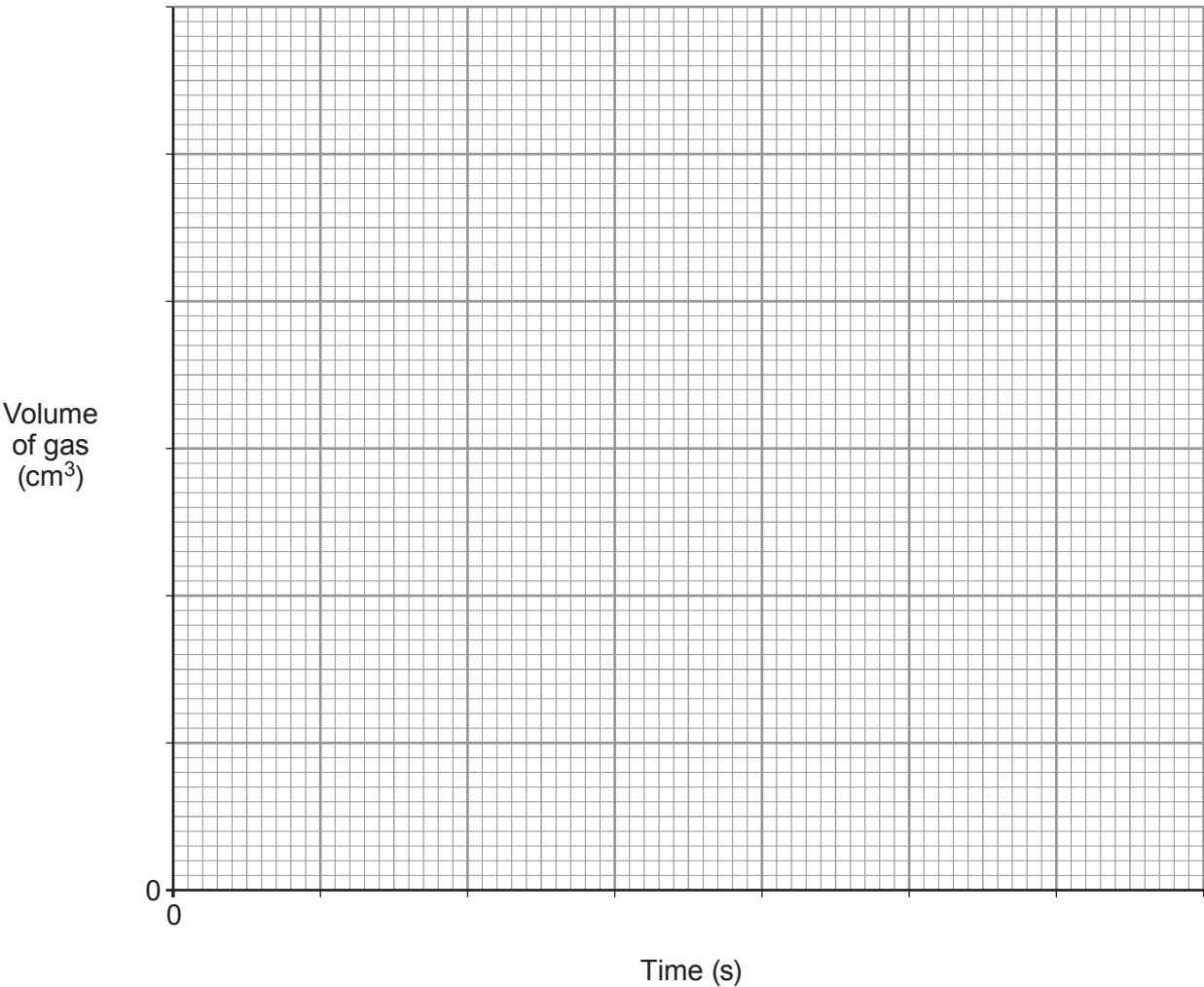
The table shows the results obtained in the experiment.

Time (s)	Volume of gas (cm^3)
0	0
10	34
20	64
30	78
40	94
50	98
60	100
70	100

- (a) Plot the data **on the grid opposite** and draw a suitable line. Label the graph **A**. [3]
- (b) **On the same grid**, sketch the graph that would be obtained if a piece of magnesium ribbon of half the length were reacted with another 60 cm^3 of the same hydrochloric acid. Label this graph **B**. [2]



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(c) If the original experiment were repeated at a higher temperature, the reaction would occur at a higher rate. Explain this using particle theory. [2]

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(d) This method can be used to investigate the rate of this reaction. Without changing the method or apparatus used, state what should be done to improve the overall strength of the evidence collected. [2]

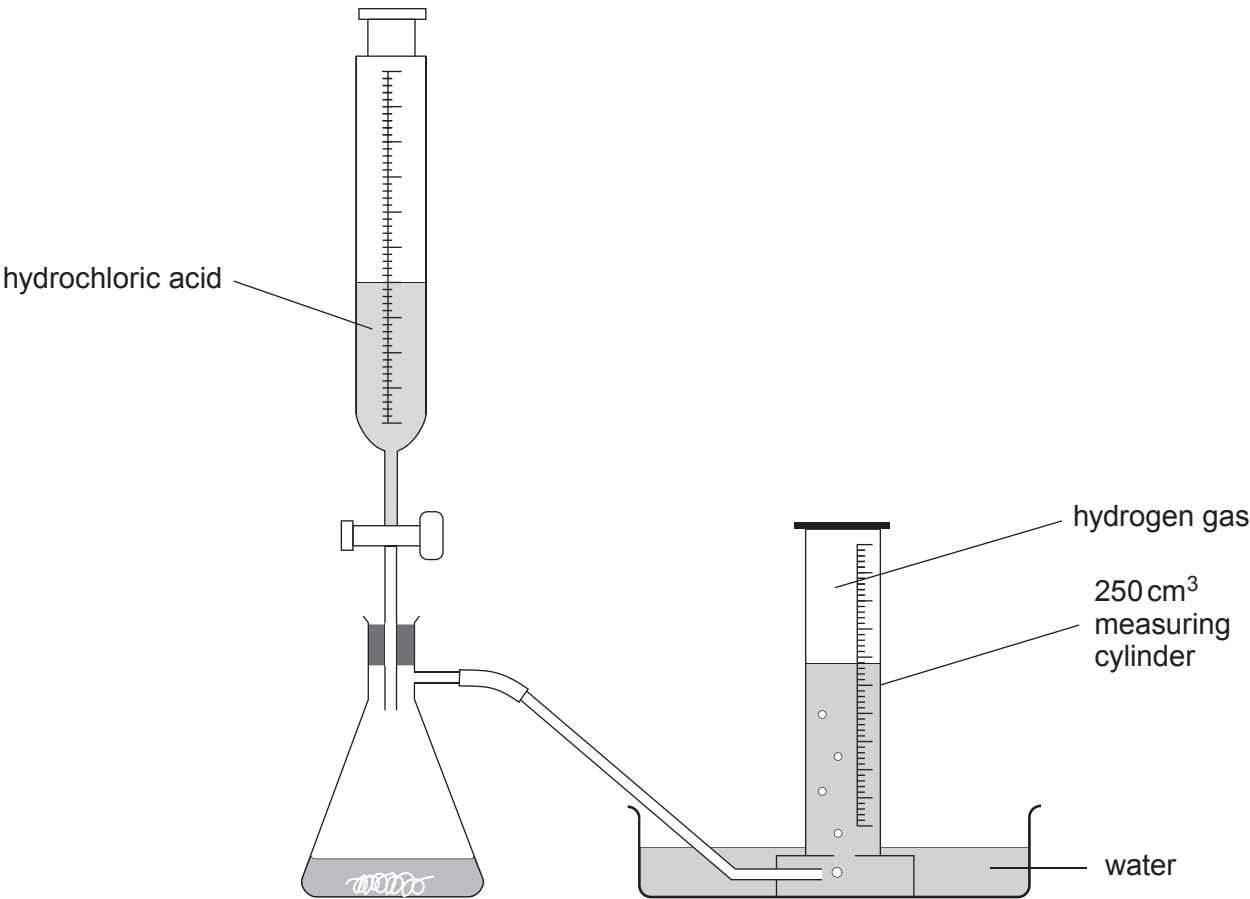
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(e) Give an advantage and a disadvantage of using the following alternative apparatus to collect the hydrogen gas. [2]



Advantage

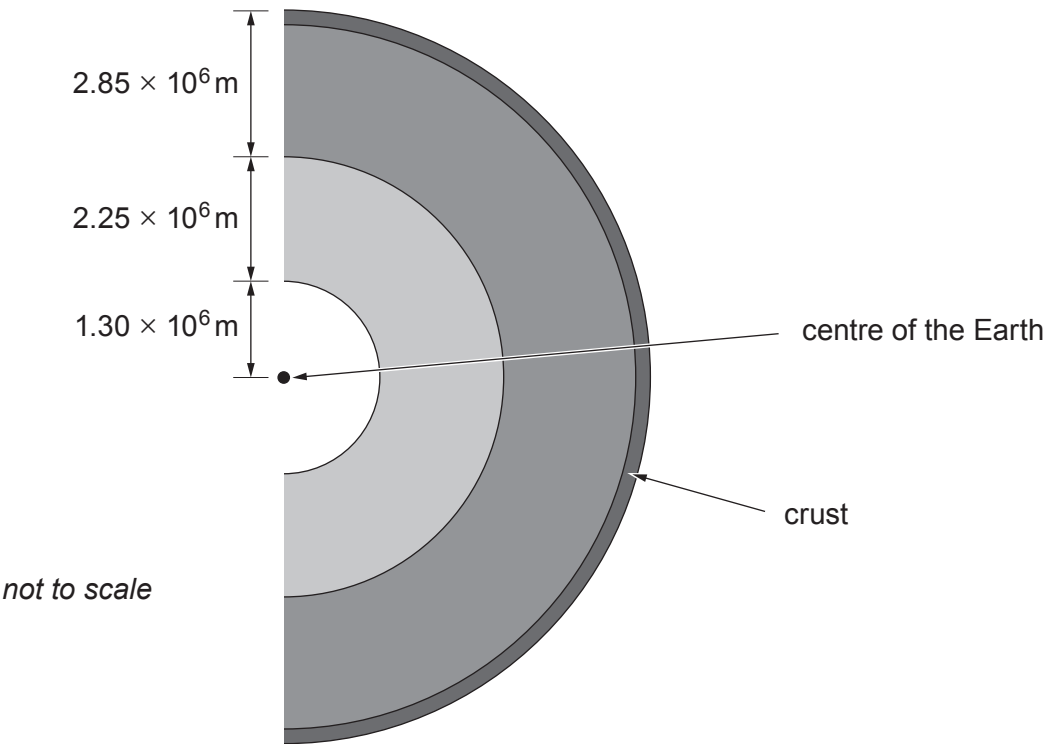
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Disadvantage

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4. The Earth is almost spherical. The diagram shows a section of the layered structure of the Earth.



(a) The thickness of the crust at one location is 0.97 % of the distance from the surface to the centre of the Earth.

Use this information to calculate the thickness of the crust. Give your answer in **standard form**. [3]

Thickness of crust = m



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(b) (i) Describe the theory of plate tectonics.

[3]

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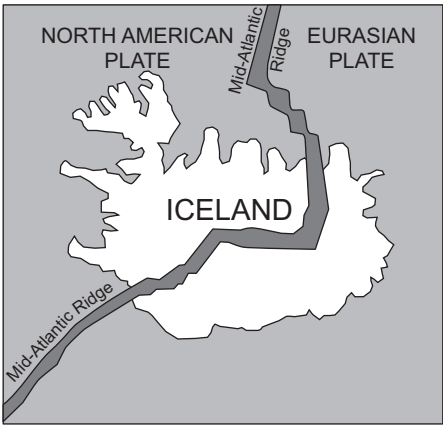
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(ii) Iceland lies on the Mid-Atlantic Ridge which is formed at a constructive plate boundary. Explain the formation of the Mid-Atlantic Ridge. [2]



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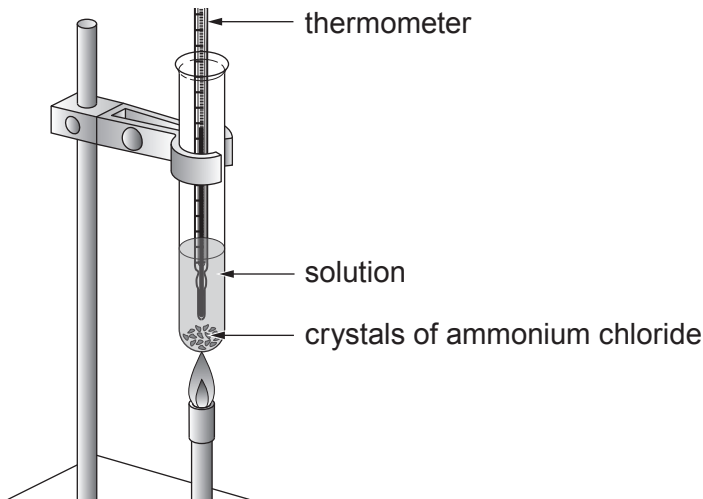
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4. A student investigates the solubility of ammonium chloride by adding different masses to 10 g of water.

He uses the apparatus shown.



10 g of water is placed in a boiling tube and 3.0 g of ammonium chloride is added.

The tube is heated until all the solid dissolves.

The tube is allowed to cool.

The temperature at which solid ammonium chloride first appears is recorded.

The experiment is repeated using different masses of ammonium chloride.

The results are shown in the table.

Mass of ammonium chloride in 10 g of water (g)	3.0	3.3	4.1	5.2	5.9	6.6
Temperature at which solid ammonium chloride first appears (°C)	4	10	30	52	68	80

(a) What practical problem is the student likely to come across in finding the first two results?
Suggest how this problem might be overcome. [2]

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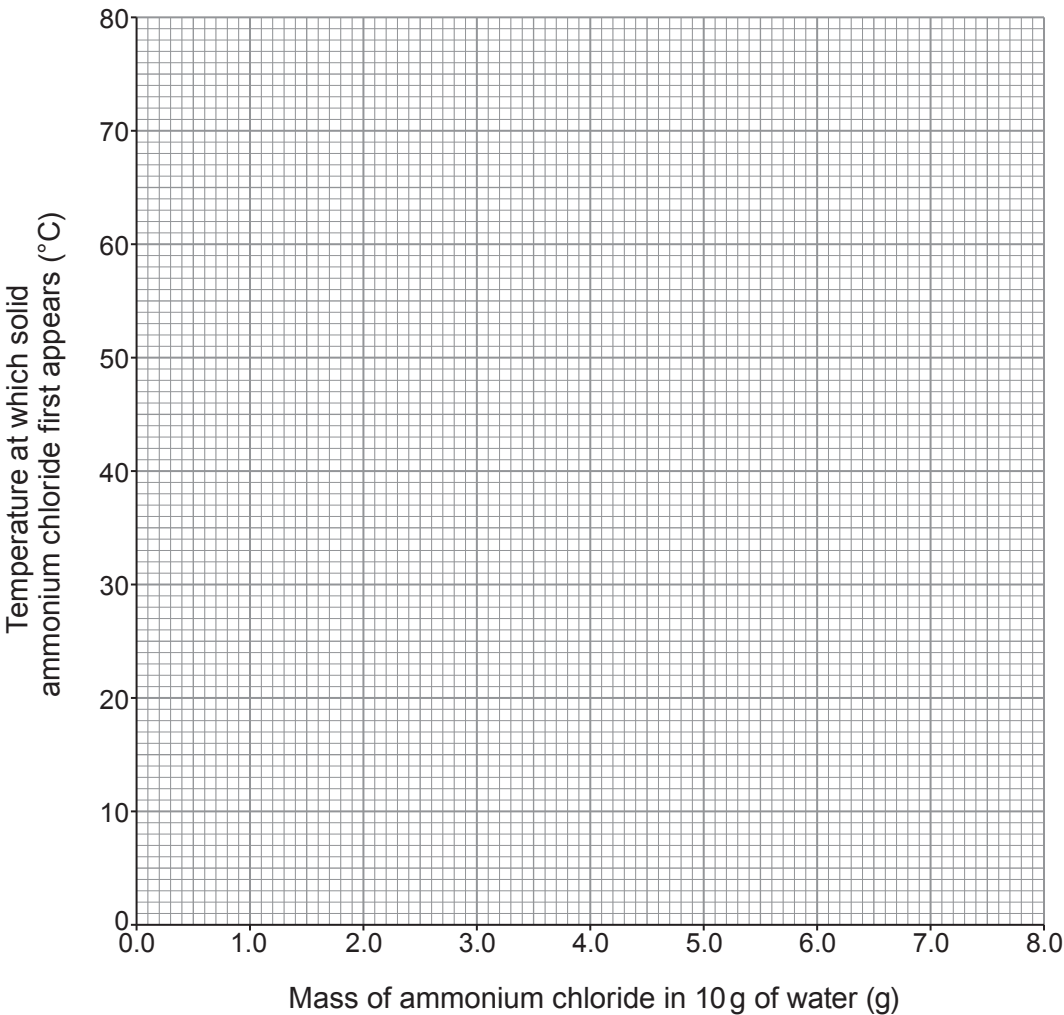
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- (b) (i) On the grid below, plot the temperature at which solid ammonium chloride first appears against the mass of ammonium chloride in 10 g of water. Draw a suitable line. [3]



- (ii) The student is given a boiling tube containing 5.0 g of ammonium chloride in 10 g of water. He stirs the ammonium chloride in the water and heats it to a temperature of 45 °C.

State whether all the ammonium chloride dissolves. Give a reason for your answer. [1]

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- (c) The student is asked to use a **different** method to find the exact solubility of another compound in water at room temperature. He knows that it has a value of approximately 7 g per 100 g of water at this temperature.

He is given a 5.0 g sample of the compound and common laboratory equipment but **no heating apparatus**.

Describe how he would carry out his method and how he would find the solubility. [3]

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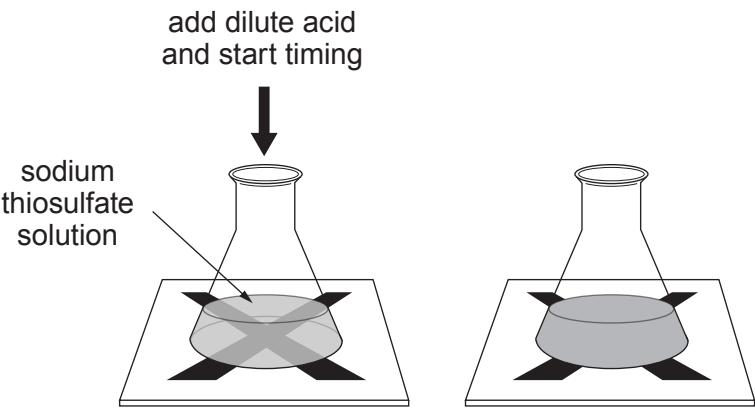
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9



4. Sodium thiosulfate solution reacts with dilute hydrochloric acid to form a precipitate. The precipitate causes the solution to go cloudy.
- The rate of the reaction can be measured by placing a cross beneath the flask and measuring the time taken for the cross to disappear.



Gareth and Sion studied the effect of sodium thiosulfate concentration by carrying out the reaction with thiosulfate of five different concentrations. They tested each concentration three times.

Their results are shown in the table below.

Concentration of sodium thiosulfate (g/dm ³)	Time 1 (s)	Time 2 (s)	Time 3 (s)	Mean time (s)
0.2	114	113	112	113
0.4	74	70	72	72
0.6	40	38	57	39
0.8	21	23	22	22
1.0	14	16	15	15

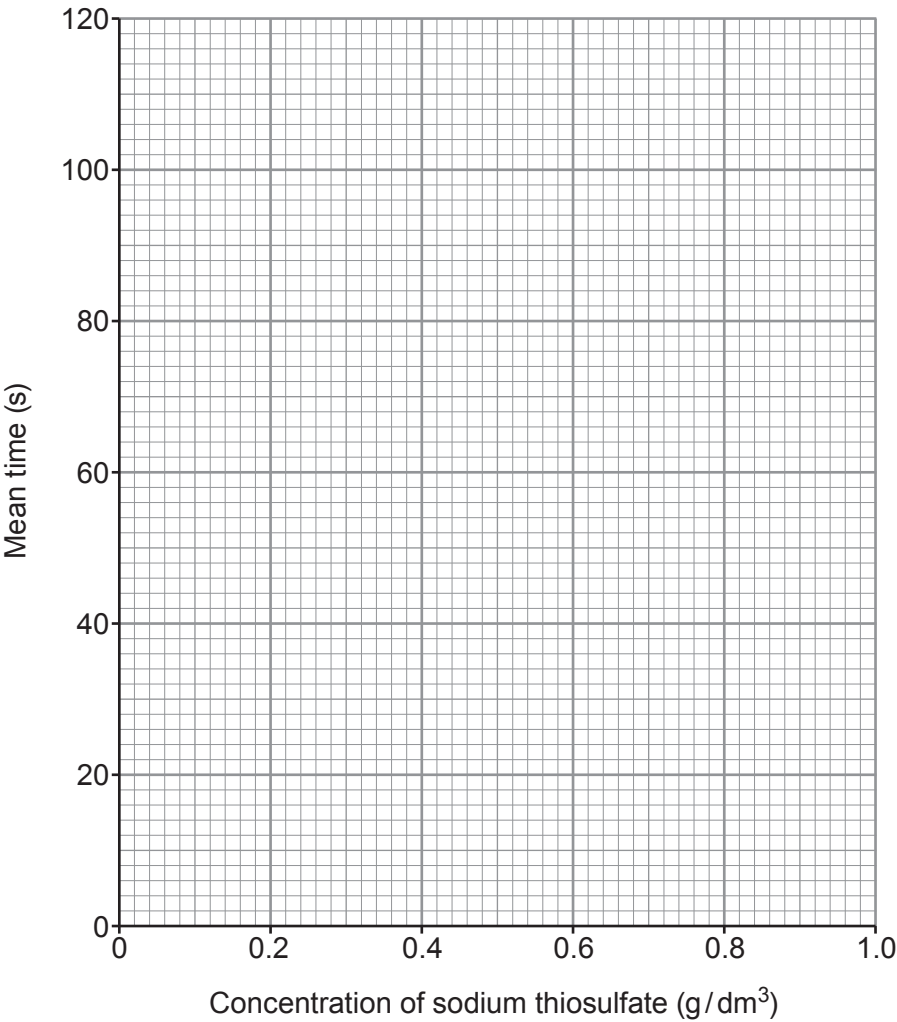
- (a) When calculating the mean times, they ignored one of the values recorded.
- Circle this value in the table.

[1]



- (b) Plot the mean time against sodium thiosulfate concentration on the grid below. Draw a suitable line.

[3]



- (c) Explain the results in terms of particle theory.

[3]

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- (d) A second group of students carried out the same experiment with sodium thiosulfate solution at a higher temperature.

On the grid above, sketch a line to show the results you would expect them to record.

[1]



- (e) The stock solution of sodium thiosulfate used in both experiments was made by dissolving 1.0 g of the solid in 1 dm³ of water.

Calculate the number of moles of sodium thiosulfate (Na₂S₂O₃) in 1.0 g.

Give your answer to **two** significant figures.

[2]

$$A_r(\text{Na}) = 23 \quad A_r(\text{S}) = 32 \quad A_r(\text{O}) = 16$$

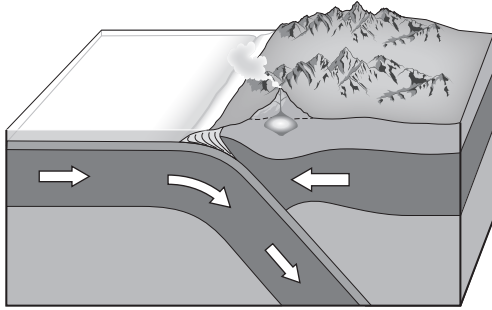
Number of moles = mol

10

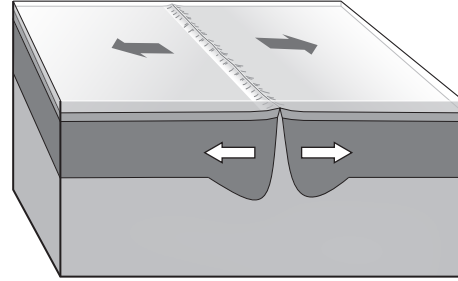


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4. The diagrams show two types of plate boundaries, **A** and **B**. Describe what happens at each boundary. [6 QER]



A



B

